

CYBR-2102 Milestone 3 — Incident Response Plan

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Course: CYBR-2102 Network Defense Fundamentals

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1 Incident Overview

Field	Entry (Example)
Incident Title	Unauthorized RDP Brute Force Attempt
Date / Time Detected	12/03/2024 – 09:12 AM
Detected By	☒ SOC Tool ☐ User ☐ Administrator
Detection Method / Source	Windows Event Log (4625 failures), Security Onion alerts
Systems Affected	WIN-SRV01 (10.0.0.15), Firewall, Domain Account “labadmin”
Severity Level (L/M/H)	Medium
Current Status	☐ Open ☒ Contained ☐ Closed

2 Incident Response Team Structure

Role	Name / Responsibility	Contact Info
Incident Manager	Jordan Smith – Oversees IR actions, approves containment	jsmith@company.com

Technical Lead	Conducts log review, validates indicators
Communications Officer	Internal updates and stakeholder communication
Legal / Compliance	Ensures regulatory alignment
HR / Management Liaison	Reports to leadership, manages personnel issues

3 Response Phases – NIST SP 800-61 Framework

Phase	Actions Taken	Evidence / Verification	Framework Reference
Preparation	Verified logging, confirmed baseline firewall rules, ensured IR team contacts updated	Screenshot of logging settings	CIS 8; NIST IR §3.1
Detection & Analysis	SOC alerted multiple failed RDP attempts; analyst validated Event ID 4625; IP reputation checked	Event Viewer logs; SIEM alert	CIS 6; NIST IR §3.2
Containment	Disabled compromised user account; added firewall block for source IPs; enforced lockout policy	Firewall rule screenshot; AD user disable confirmation	CIS 4, 5, 13; NIST IR §3.3
Eradication	Checked system for malware; removed unauthorized scheduled task; validated integrity of RDP config	AV scan results; Task Scheduler screenshot	CIS 10; NIST IR §3.4
Recovery	Re-enabled secure admin account; implemented stronger password policy; monitored for 48 hours	Password policy screenshot; monitoring logs	CIS 4, 5; NIST IR §3.5
Post-Incident Activity	Conducted lessons learned meeting; documented RCA and recommendations; updated IR playbook	Meeting summary; final IR report	CIS 17; NIST IR §3.6

4 Timeline of Events

Time Stamp	Event Description	Action Taken	Responsible Party
09:12 AM	Failed RDP login attempts detected	Flagged by SOC	Analyst
09:14 AM	Confirmed multiple source IPs	Log review initiated	SOC Lead
09:18 AM	Account disabled / Firewall rule added	Containment complete	Admin
09:30 AM	Malware scan executed	System validated	Technical Lead
10:05 AM	Password policy updated	Enforced complexity and lockout	SysAdmin
11:00 AM	Monitoring period initiated	48-hour watch set	Incident Manager

(Add or modify entries to match your lab timeline.)

5 Root Cause Analysis & Findings

Summary:

The incident was caused by external attackers attempting to brute-force RDP credentials. *A weak password and lack of account lockout policy* allowed repeated attempts without immediate lockout enforcement.

Contributing Factors:

- RDP exposed to the internet
 - Inadequate password complexity
 - No geo-blocking firewall rules
 - Missing alert thresholds for repeated failed logins
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6 Recommendations / Preventive Actions

Control Area	Recommended Action	Priority	Verification Step
Access Control	Enforce MFA for all remote connections	High	Successful MFA test login
Patch Management	Ensure monthly patch cycle applied to RDP servers	Medium	Confirm patch status via WSUS/Windows Update
Firewall Rules	Block RDP from external networks; implement geo-blocking	High	Validate rule in firewall console
User Awareness Training	Conduct password hygiene and social engineering training	Medium	Completion certificates; quiz results

7 Evidence Checklist

- ☒ Security Log screenshots
 - ☒ Firewall update evidence
 - ☒ IR Plan table complete
 - ☒ Timeline filled
 - ☒ Recommendations included
 - ☒ File named correctly (Lastname_CYBR2102_M3.pdf)
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8 Sign-Off

Step	Completed By	Date	Verified By (Instructor)
Incident Response Plan Created			
Evidence Attached			
Recommendations Documented			